
Chapter 14

Human Communication



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✓ Learning Objectives

Language Production and Comprehension: Brain Mechanisms

- LO 14.1** Compare the language-related functions of the left and right hemispheres, and discuss the prevalence of these functions in the opposite hemisphere.
- LO 14.2** List factors involved in language production, and identify corresponding brain regions.
- LO 14.3** Describe the factors involved in language comprehension, and identify corresponding brain regions.
- LO 14.4** Describe evidence for common and language-specific brain regions for bilingual language processing.
- LO 14.5** Identify brain structures and functions involved in prosody.
- LO 14.6** Identify the brain regions involved in recognizing people's voices.

Disorders of Language Production and Comprehension

- LO 14.7** Describe the symptoms and neural mechanisms of Broca's aphasia, including agrammatism, anomia, and articulation.
- LO 14.8** Describe the symptoms and neural bases of Wernicke's aphasia, including deficits in spoken word recognition, comprehension, and converting thoughts into words.
- LO 14.9** Identify the symptoms of conduction aphasia, and describe their contribution to understanding about connections between Broca's and Wernicke's areas.
- LO 14.10** Explain the brain processes involved in production and comprehension of American Sign Language.
- LO 14.11** Describe the biological basis for stuttering and identify treatment strategies.

Disorders of Reading and Writing

- LO 14.12** Compare aphasias and disorders of reading and writing.
- LO 14.13** Identify the symptoms of pure alexia, and describe the brain structures involved.
- LO 14.14** Describe how research on acquired and developmental dyslexia helps us to understand the role of the brain in reading.
- LO 14.15** Explain how research on audition, vision, memorization, and motor control helps us to understand the role of the brain in writing.

Chapter 15

Neurological Disorders



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Learning Objectives

Tumors and Seizures

- LO 15.1** Describe the primary symptoms, causes, and treatments for brain tumors.
- LO 15.2** Describe the primary symptoms, causes, and treatments for seizures.

Cerebrovascular Accidents

- LO 15.3** Explain how cerebrovascular accidents can occur.
- LO 15.4** Explain how treatments can be used to address the immediate and long-term symptoms of cerebrovascular events.

Traumatic Brain Injury

- LO 15.5** Identify some causes of traumatic brain injury.
- LO 15.6** Describe treatments for traumatic brain injuries.

- LO 15.14** Describe the symptoms, causes, and treatments for multiple sclerosis.
- LO 15.15** Describe the symptoms, causes, and treatments for Alzheimer's disease.
- LO 15.16** Describe the symptoms and causes of Korsakoff's syndrome.

Disorders of Development

- LO 15.7** Describe the effects of alcohol on development of the nervous system.
- LO 15.8** Contrast the symptoms, causes, and treatments of inherited metabolic disorders.
- LO 15.9** Identify the cause, symptoms, and interventions for Down syndrome.

Degenerative Disorders

- LO 15.10** Describe how transmissible spongiform encephalopathies cause brain damage.
- LO 15.11** Describe the symptoms, causes, and treatments for Parkinson's disease.
- LO 15.12** Describe the symptoms, causes, and treatments for Huntington's disease.
- LO 15.13** Describe the symptoms, causes, and treatments for amyotrophic lateral sclerosis.

Disorders Caused by Infectious Diseases

- LO 15.17** Identify the symptoms, causes, and treatments for encephalitis.
- LO 15.18** Identify the symptoms, causes, and treatments for meningitis.

Chapter 16

Schizophrenia, Affective Disorders, Anxiety Disorders, and OCD



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Learning Objectives

Schizophrenia

- LO 16.1** Describe the positive, negative, and cognitive symptoms of schizophrenia.
- LO 16.2** Describe genetic factors associated with the development of schizophrenia.
- LO 16.3** Describe environmental factors associated with the development of schizophrenia.
- LO 16.4** Provide evidence of the role of the mesolimbic dopamine pathway in the positive symptoms of schizophrenia.
- LO 16.5** Provide evidence of the role of cortex in the negative and cognitive symptoms of schizophrenia.

Affective Disorders

- LO 16.6** Contrast the symptoms of major depressive disorder and bipolar disorder.

Anxiety Disorders

- LO 16.14** List the symptoms of anxiety disorders.
- LO 16.15** Describe factors associated with the development of anxiety disorders.
- LO 16.16** Summarize pharmacological treatments for anxiety disorders.

- LO 16.7** Describe evidence that supports a genetic contribution to the development of affective disorders.

- LO 16.8** List the types of biological treatments for affective disorders.

- LO 16.9** Describe the role of the frontal cortex in depression.

- LO 16.10** Summarize the evidence in support of the monoamine theory of affective disorders.

- LO 16.11** Critique the evidence for the role of the promoter region of the 5-HT transporter gene in affective disorders.

- LO 16.12** Explain the relationship between affective disorders, their treatment, and neurogenesis.

- LO 16.13** Summarize the involvement of circadian rhythms in affective disorders.

Obsessive-Compulsive Disorder

- LO 16.17** List the symptoms of OCD.

- LO 16.18** Describe the roles of genetic and environmental factors in the development of OCD.

- LO 16.19** Summarize treatments for OCD.

Chapter 17

Neurodevelopmental Disorders, Stress, Posttraumatic Stress Disorder, and Substance Use Disorders



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✓ Learning Objectives

Neurodevelopmental Disorders

- LO 17.1** Describe the symptoms and possible causes of autism spectrum disorder.
- LO 17.2** Describe the symptoms and possible causes of ADHD.

Stress

- LO 17.3** Compare the SAM system and HPA axis in coordinating a stress response.
- LO 17.4** Describe the negative health outcomes associated with chronic stress.
- LO 17.5** Compare the effects of long-term glucocorticoid exposure and early nurturing experiences on the brain in response to stress.

- LO 17.6** Summarize the relationship between stress and infectious disease.

Posttraumatic Stress Disorder

- LO 17.7** List the symptoms of PTSD.
- LO 17.8** Describe the roles of genetic and environmental factors in the development of PTSD.
- LO 17.9** Summarize treatments for PTSD.

Substance Use Disorders

- LO 17.10** Describe how positive and negative reinforcement contribute to the development of substance use disorders.
- LO 17.11** Describe commonly abused drugs.
- LO 17.12** Summarize the research on effective treatment for substance use disorders.